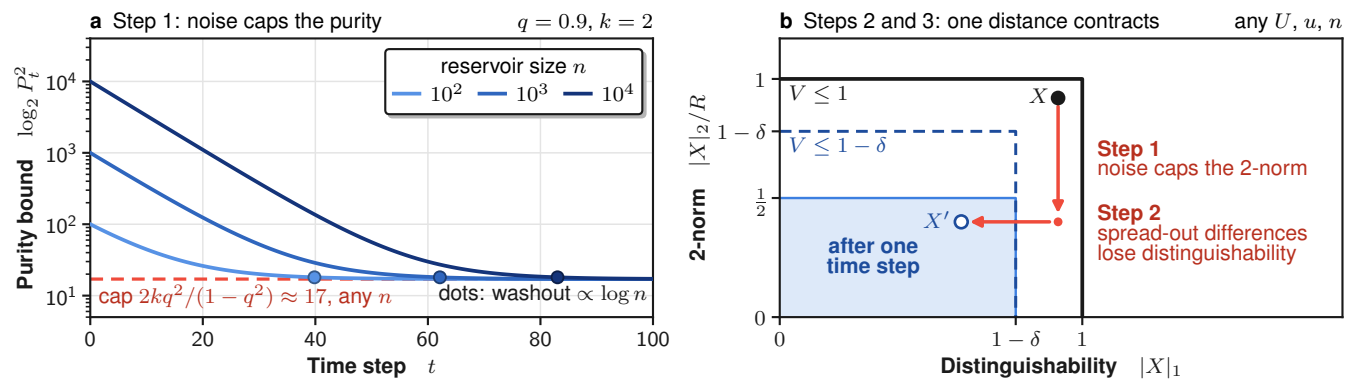

a Step 1: noise caps the purity
$q = 0.9$, $k = 2$
reservoir size n
10^2 10^3 10^4
Purity bound $\log_2 P_t^2$
10^4
10^3
10^2
10^1
cap $2kq^2/(1-q^2) \approx 17$, any n
dots: washout $\propto \log n$
0 20 40 60 80 100
Time step t
b Steps 2 and 3: one distance contracts
any U, u, n
2-norm $|X|_2/R$
1
$1-\delta$
$\frac{1}{2}$
0
$V \le 1$
$V \le 1-\delta$
X
X'
after one time step
Step 1
noise caps the 2-norm
Step 2
spread-out differences lose distinguishability
0
$1-\delta$
1
Distinguishability $|X|_1$